Insertion at beginning

```c
#include <stdio.h>
#include <stdlib.h>

struct node{
    int data;
    struct node *prev;
    struct node *next;
};

struct node *head = NULL;

void beginning() {
    struct node * first,*second,*last;

    first = (struct node*)malloc(sizeof(struct node));
    second = (struct node*)malloc(sizeof(struct node));
   last = (struct node*)malloc(sizeof(struct node));

    first->data = 10;
    second->data = 20;
  last->data = 30;

    head =  first;

     first->next = second;
    second->prev =  first;
```

```
After inserting: 1 10 20 30

=== Code Execution Successful ===
```

```c
    second->prev =  first;

    second->next =last;
last->prev = second;

   last->next = head;
    head->prev =last;
}

void insertStart(int data){
    struct node *newnode,*last;

    newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = data;

  last = head->prev;

    newnode->next = head;
    newnode->prev =last;

    last->next = newnode;
    head->prev = newnode;

    head = newnode;
}

void display(){
```

```
After inserting: 1 10 20 30

=== Code Execution Successful ===
```

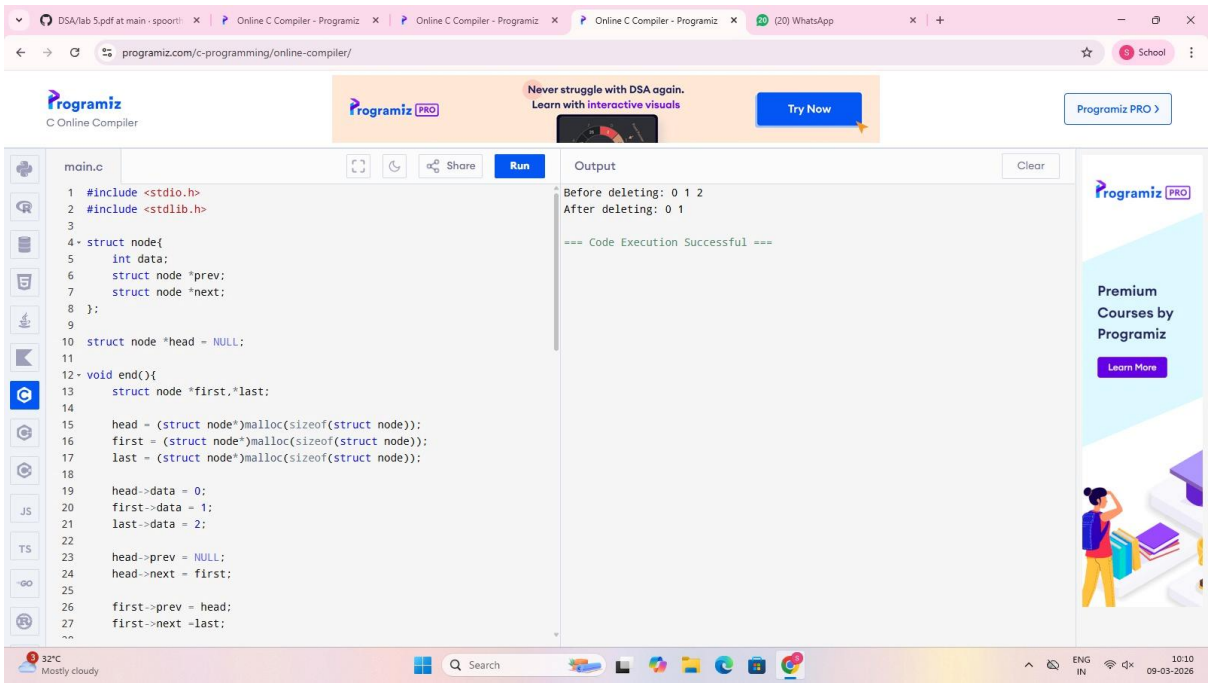
main.c

```c
void display(){
    struct node *temp = head;

    if(head == NULL){
        printf("List is empty");
        return;
    }

    printf("%d ", temp->data);
    temp = temp->next;

    while(temp != head){
        printf("%d ", temp->data);
        temp = temp->next;
    }
}

int main(){
    beginning();
    insertStart(1);

    printf("After inserting: ");
    display();

    return 0;
}
```

Output

```
After inserting: 1 10 20 30

=== Code Execution Successful ===
```

Deletion at End



DSA/lab 5.pdf at main · spoorthi | Online C Compiler - Programiz | Online C Compiler - Programiz | Online C Compiler - Programiz | (20) WhatsApp
programiz.com/c-programming/online-compiler/
Programiz C Online Compiler
Programiz PRO Learn Data Structures and Algorithms with live code visualizations Try Now
Programiz PRO

main.c

```c
    first->prev = head;
    first->next =last;

    last->prev = first;
    last->next = NULL;
}

void deleteEnd(){
    struct node *delNode = head;

    if(head == NULL){
        printf("List is empty\n");
        return;
    }

    while(delNode->next != NULL){
        delNode = delNode->next;
    }

    if(delNode->prev != NULL){
        delNode->prev->next = NULL;
    }
    else{
        head = NULL;
    }

    free(delNode);
```

Output

```
Before deleting: 0 1 2
After deleting: 0 1

=== Code Execution Successful ===
```

Programiz PRO
Premium Courses by Programiz
Learn More

32°C Mostly cloudy
Search
ENG IN 10:11 09-03-2026

main.c

```c
    }

    free(delNode);
}

void display(){
    struct node *temp = head;

    while(temp != NULL){
        printf("%d ", temp->data);
        temp = temp->next;
    }
}

int main(){
    end();

    printf("Before deleting: ");
    display();

    deleteEnd();

    printf("\nAfter deleting: ");
    display();

    return 0;
}
```

Output

```
Before deleting: 0 1 2
After deleting: 0 1

=== Code Execution Successful ===
```

Deletion at position

```c
#include <stdio.h>
#include <stdlib.h>

struct node{
    int data;
    struct node *prev;
    struct node *next;
};

struct node *head = NULL;

void insertAtposition(){
    struct node *first, *last;

    head = (struct node*)malloc(sizeof(struct node));
    first = (struct node*)malloc(sizeof(struct node));
    last = (struct node*)malloc(sizeof(struct node));

    head->data = 10;
    first->data = 20;
    last->data = 30;

    head->prev = NULL;
    head->next = first;

    first->prev = head;
    first->next = last;
```

```
Before deletion: 10 20 30
After deletion: 10 30

=== Code Execution Successful ===
```

```c
    last->prev = first;
    last->next = NULL;
}

void deletePos(int pos){
    struct node *delNode = head;
    int i;

    if(head == NULL){
        printf("List is empty\n");
        return;
    }

    if(pos == 1){
        head = head->next;
        if(head != NULL)
            head->prev = NULL;
        free(delNode);
        return;
    }

    for(i = 1; i < pos-1; i++){
        delNode = delNode->next;
    }

    struct node *del = delNode->next;
```

```
Before deletion: 10 20 30
After deletion: 10 30

=== Code Execution Successful ===
```

programiz.com/c-programming/online-compiler/

Programiz C Online Compiler

main.c

```c
        head->prev = NULL;
        free(delNode);
        return;
    }

    for(i = 1; i < pos-1; i++){
        delNode = delNode->next;
    }

    struct node *del = delNode->next;

    delNode->next = del->next;

    if(del->next != NULL)
        del->next->prev = delNode;

    free(del);
}

void display(){
    struct node *temp = head;

    while(temp != NULL){
        printf("%d ", temp->data);
        temp = temp->next;
    }
}
```

Output

```
Before deletion: 10 20 30
After deletion: 10 30

=== Code Execution Successful ===
```

programiz.com/c-programming/online-compiler/

Programiz C Online Compiler

main.c

```c
    free(del);
}

void display(){
    struct node *temp = head;

    while(temp != NULL){
        printf("%d ", temp->data);
        temp = temp->next;
    }
}

int main(){

    insertAtposition();

    printf("Before deletion: ");
    display();

    deletePos(2);

    printf("\nAfter deletion: ");
    display();

    return 0;
}
```

Output

```
Before deletion: 10 20 30
After deletion: 10 30

=== Code Execution Successful ===
```